

T9 (Abel) If a power series $\sum a_n x^n$ converges at a point $x_0 \neq 0$, then it converges uniformly and absolutely in every closed disk.

$$\bar{D}(0, r) = \{x \in \mathbb{C} \mid |x| \leq r\} ; 0 < r < |x_0|$$

Proof: Let $x \in D$. Since the series $\sum a_n x_0^n$ is convergent, there exists a number M such that.

$$|a_n x_0^n| \leq M ; (\forall n \in \mathbb{N}) \quad (\text{majorant})$$

We have:

$$\sum |a_n x^n| = \sum \underbrace{|a_n x_0^n|}_{\frac{x}{x_0} \text{ in } n} \left| \frac{x}{x_0} \right|^n \leq M \sum \underbrace{\left| \frac{x}{x_0} \right|}_{< 1}^n, \quad (\forall x \in D)$$

for all $x \in D$. By Weierstrass's theorem, the series $\sum a_n x^n$ is uniformly and absolutely convergent in D . ✓

We use the notation

$$\overline{\lim} a_n = \sup \{ \text{the limit points of } a_n \}$$

limita superiora supremum

$$\overline{\lim} (-1)^n = 1,$$

$$\overline{\lim} \frac{(-1)^n}{n} = 0;$$

$$\overline{\lim} (1 + (-1)^n)^n = \infty.$$

D10. The radius of convergence of a power series is the radius of the largest disk in the interior of which the series converges. It is either a non-negative real number or ∞ .

T11. The element

$$R = \frac{1}{\overline{\lim} \sqrt[n]{|a_n|}} \in [0, \infty)$$

on condition $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$, is the radius of convergence of the series $\sum a_n x^n$.

The set

$$D(0, R) = \{x \in \mathbb{C} \mid |x| < R\}$$

is the disc of convergence of the series.